



ROLE AND IMPORTANCE OF KEY PERFORMANCE INDICATORS MEASUREMENT

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Abstract

Key performance indicators are financial and non financial indicators that organizations use in order to estimate and fortify how successful they are, aiming previously established long lasting goals. Appropriate selection of indicators that will be used for measuring is of a greatest importance. Process organization of business is necessary to be constitute in order to realize such effective and efficient system or performance measuring via KPI. Process organization also implies customer orientation and necessary flexibility in nowadays condition of global competition.

Explanation of process organization, the way of KPI selection, and practical example of KPI measuring in Toyota dealerships are presented in this paper.

Keywords: key performance indicators (KPI), process approach, Toyota dealerships

1. INTRODUCTION

New business conditions where information is the most important resource impose new approaches in measuring performances of organizations, related to traditional performance measurement system which evolved just financial and accounting indicators. One of the newer approaches

refer on measuring performances of organizations via KPI. KPI are financial and non- financial measures that organizations use to reveal how successful they were in accomplishing long lasting goals. In order to constitute effective system of performance measurement it is very important to have defined and standardized all processes within the organization.

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Significance of process approach could be seen through the fact that it is a base of two nowadays management systems: strategic management, via BSC, and technical standardization, via QMS.

Due to that fact in first part of this paper will be explained the meaning of process approach, and a way of constituting it. In second part of this paper will be explained what are KPI, and importance of organization performances measurement, and in third part of the paper will be explained and shown some of the KPI that Toyota dealerships uses in their business.

2. PROCESS APPROACH

As it was previously said constitution of process organizations is a necessary base for constitution of measurement system via KPI. Such statement could also be find in the literature: „Business process are base for organization functioning because companies are constituted basically of processes, not products or services. On the other words, business management of some organizations means process management“ (Skrinjar et al., 2007; Stefanovic et al., 2010).

Process definitions are base on which the whole philosophy of organization functioning have been constituted. The process approach means that attention is shifted from end output (products and services) to the activity chain that shapes this output (Rentzhog, 1997). Importance of process orientation arise form the first half of XX century, where Walter A. Shewharts stated that highly qualified product could be get only by process management.

Process orienatation constitutions are very important. Traditional ways of business that imply functional and hierarchical

approach is not enough good in nowadays conditions, where rapid changing of business conditions seeks flexibility. Traditional organizations are too rigid to those changes. Process approach enables organization to steer their business to main duty, which is to make values that will satisfy their customers.

Advantage of process organizations are (Heleta, 1998):

1. Put customers claims at the first place,
2. Constitute interfunctional management,
3. Defined process owners,
4. Activities are done logically,
5. Employees do just they need to do,
6. Internal relation suppliers - customer are constituted.

Toyota dealerships whose control of effectiveness and efficiency of business, via KPI, will be presented in this paper, are constituted as process organizations. In such organization, all processes are prescribed and standardized. The hierarchical levels of defined processes are given in figure 1.

Descriptions of those levels are defined as follows:

1. Toyota customer oriented process (TCOP) are essential part of activities at dealership; and every action have a significant role in accomplishing perfect situation within TCOP.
2. Elements of Toyota operation activities (TO) like people, object and internal procedures that are focused to accomplish TCOP
3. Business management of after-sales activities (MGMT) and performance measurement system put appropriate attention to operation management.

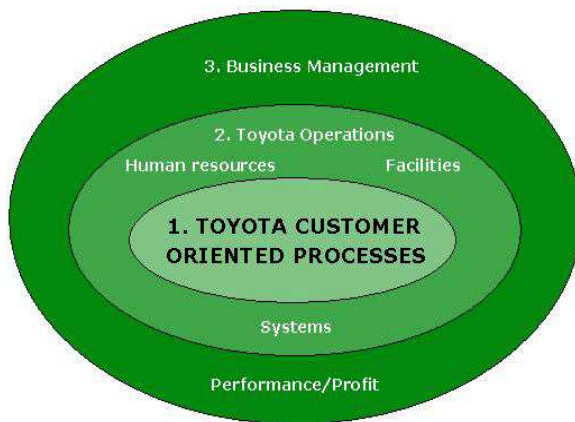


Figure 1. Hierarchical levels of Toyota processes (Radojevic et al., 2009)

3. ROLE AND IMPORTANCE OF MEASURING ORGANIZATIONAL PERFORMANCES VIA KPI

Previously described process organization is a necessary base for KPI defining. Such system of standardized activities allows adequate measuring of performances. Importance of a measuring is significant. Continual measuring is a base for continual improvements of organization performances that is one of the most important management principle (Besic & Djordjevic, 2007). One proverb may be found in literature: „If you want to improve something, you have to measure it“ (Radovic & Karapandzic, 2005).

An effective organization knows that if they don't have enough informations about process, product or service, they can't control that part. There are large numbers of values measured during work of some organization. Lord Kelvin, British physician, in 1891, spoke: „When you can measure something that you are talking about, you know something about it“.

To stay competitive, organization should manage with employees, processes, planned

activities, reductions times, relations with suppliers, and other parts of the business. System for effective measuring of performances is used to understand, adjust and improve business in all department of the organization (Summers, 2005).

Measuring performances of the organization means qualitative, and quantitative expression of some results by chosen indicators. Performance measurement enable to effective organizations to express their success by numbers. Selection of appropriate indicators that will be used for measurement and appraisal of the performances is a very important activity. Among all informations that can be get it is necessary to choose some critical quantity that on the best way represent the whole business.

Beside control function indicators of performances also have two next functions:

- Developing and guiding function – because they present a base for formulating and implementation of the strategy of the organization,
- Motivation function – induce management to fulfill goals and motivate all stakeholders to realize those goals and on even higher level (Pesalj, 2006; Stamatović & Zakić, 2010).

In all organizations, an employee knows that there are activities that are very important for the management team. In sense of defining a control package of indicators that represent success of some business conception of Key performance indicators were appeared. Key performance indicators (KPI) are financial and non-financial indicators that organization uses to testify how successful they were in achievement of long lasting goals.

KPI are static and stable indicators that carry more meaning when comparing

information. They help to remove the emotion away from object of the business, and get one focused on the thing that job is really about, and that is making profit.

4. KPI IN TOYOTA DEALERSHIPS

To ensure ease of reference, as well as comprehensive indexing, one of the most effective approach in KPI defining, is to divide the whole business into appropriate sections, which represent different parts of dealership.

These five divisions are Sales Department, Service Department, Parts Department, Body shop and Business Management. In the further text, some of the basics KPI of all of the divisions will be presented, just to show the non complexity of defining and using them. Some of the indicators that were measured in one representative Toyota dealership will be also shown.

4.1. KPI in Sales department

Vehicle Gross Profit (Invoice Price of Vehicle – Cost price of Vehicle)

The basic understanding of Gross Profit is simply sales less the cost of those sales. Within the sales department there are many things that are sold that are all contained on the sales invoice such as accessories, warranty and of course the vehicle itself. Gross profit is often expressed in percentage of vehicle sales price, for example:

Gross profit (%) = (Vehicle Sales price – Vehicle cost price) / (Vehicle Sales Price) x 100

Present economic crisis effect car dealers

the most. Quite often they struggle to retain business with positive GP. Market slowdown could influence sales even with negative GP.

Annualized sales per sales person
(Annualized unit sales/number of Sales people)

This KPI is made of Annualized unit sales. Simply, measures the average number of units a Sales person sells in 1 year. This statistic provides information of average sales team performance and clear perception for each member of the team to accurately assess the strengths and weaknesses.

General benchmark in car dealership is 150 units per annum for newcomer sales person and well established salesperson should sell to 200 units per annum.

New car sales are dropping down in years of crisis. It is important to retain Vehicle GP as well as New Car Sales in goal to achieve good business results. (Volume quantity and quality of sales).

Stock turn (Annualized unit sales / Units in stock)

Vehicle stock turn provide information of number of times that vehicle stock turnover in one year. This KPI is very important because it provide convenient monitoring of stock. The faster dealer turn vehicle stock, less money is needed for invest and that means more profit. Again this KPI tends to increase dealer profitability in vehicles and speeds circulation of funds.

In the case of low Stock turn, management should invest in training of sales person, and marketing advertisement. Comparative data about plan and sales of new cars during a period of one year are shown on figure 2.

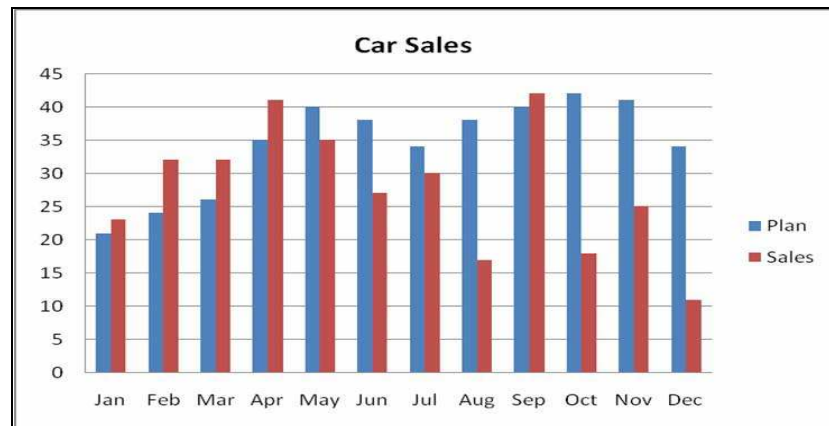


Figure 2. Volume of car sales within the period of one year

4.2. KPI in Service Department

Productive Staff:Non-Productive Staff

This KPI measures the ratio between productive staff and non-productive staff. It is also called Productive Ratio.

Typically, non-productive staff could be the Service Manager, Service Receptionist and Warranty Clerk.

Composite results appear to average at around 3:1, and that means that for every 3 Technicians that you employ, you have 1 member of staff who is non-productive. If this ratio is lower than 3:1 you will need to take corrective action.

Technical efficiency TE(%)

Technical efficiency = Hours Sold/Hours Work (%)

This KPI measures the relationship between the number of hours that have been sold and the number of hours that the Technicians have worked. Industrial European technical efficiency average is about 115%.

It is not so easy to achieve this efficiency in ordinary workshop. But it is important to strive to manage this benchmark. In case that workshop is having problems to achieve good result in TE it is good to propose several counter measures: 1) first of all it is very important that dealership has accurate measurement system for this indicator, 2) management should organize technician skill level appraisal and if it is not appropriate, they have to organize technical trainings, 3) Management have to observe condition of equipment and tools in goal to increase TE, and to propose new equipment which can upgrade efficiency, 4) management have to consider implementation of bonus system in goal to reward technicians personal achievements.

Labor Utilization LU (%)

Labor Utilization = Hours Worked/Hours Attended (x100)

This KPI is also known as Utilization, Labor efficiency and Selling efficiency. The statistic tells how much of the productive workers attended time is actually spent working productively. Benchmark for this

indicator it from 85% to 95%. Industrial European Labor Utility average is about 90%.

In case that dealerships have a poor LU problem can be induced by poor organization and bad habits as well as low quality work of receptionist who is actually selling available workshop time. Reasonable countermeasures should be suggested by management: 1) adequate skill training for reception employees, 2) considering of bonus system implementation for employs.

Overall Productivity OP (%)

Overall productivity = $\frac{\text{Hours Sold}}{\text{Hours Worked Productively}} \times 100$

This KPI is also known as Productive Efficiency or Working efficiency. It shows ability of Technicians to complete their work within the franchise manufacturers allocated time schedules.

Most franchise manufacturers provide dealers with allocated times for jobs on all vehicles and this allocated time is the maximum amount that dealer is able to

charge customer. This allocated times are written in document named Flat Rate Time Manual.

If productive workers take more time to complete the job then the manufacturers flat rate time suggest then dealers profitability diminishes.

Benchmark for this KPI is from 110 to 125%. Industrial European Overall productivity average of OP is about 104%.

Graphical example of KPI within this group of parameters within one Toyota dealership, are given on figure 3.

4.3. KPI in Parts department

Annualized parts sales

In order to calculate many KPI dealer parts sales volume may need to be annualized. The formula is the year-to-date sales volume figure, multiplied by 12 then divided by the current month number. This is a theoretical figure that provides dealer with the sales value, that dealers parts department should achieve at the end of the year if sales

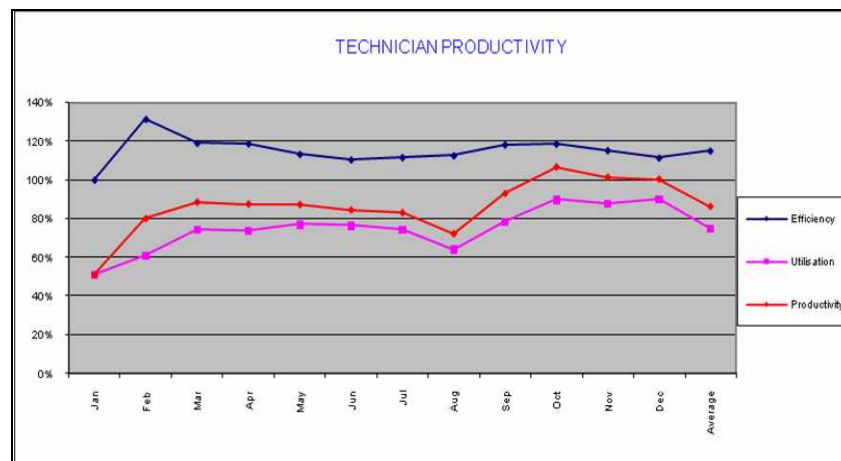


Figure 3. Workshop KPI

performances where to be maintained at the current rate.

In case that dealership face problem to achieve planned turnover, management should consider countermeasures to achieve goals. Suggested countermeasures are marketing department support, regarding spare parts sales, spare parts price analyses in goal to challenges competitiveness.

Parts gross profit (%)

Parts gross profit = $\frac{\text{Gross profit}}{\text{Departmental Turnover}} \times 100$

Financial reports for the Parts Department show Gross Profit on each individual category of Parts Sales as well as the overall total parts Sales.

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It is essential to compare parts gross profit with dealership gross profit to ensure that business is traveling in good direction.

Figure 4 shows measured sales volume of Spare Parts in mentioned Toyota dealership.

4.4. KPI in Body shop

Labor Cost of Sales

Labor Cost of Sales = Value of Hours Attended – Idle time

This statistic captures the amount of money that is paid to Productives for every hour that is worked productively. The amount of money that dealerships pay their Productives per hour is divided into three sections: Hours Attended, Hours Worked Productively, and Idle time.

Hours Attended simply represents the number of hours that the Productives have been clocked in and available to work, these hours are simply divided between Hours Worked Productively and Idle Time. Value of this KPI shouldn't be less then 90% of Attended Hours.

Labour Sales Mix

Labour Sales Mix = $\frac{(\text{Sector}) \text{ Hours Sold}}{\text{Total hours sold}} \times 100$

This information informs dealer of the sectors into which hours are sold. Labour sales mix for the body shop in split into three distinct categories of income sectors, which

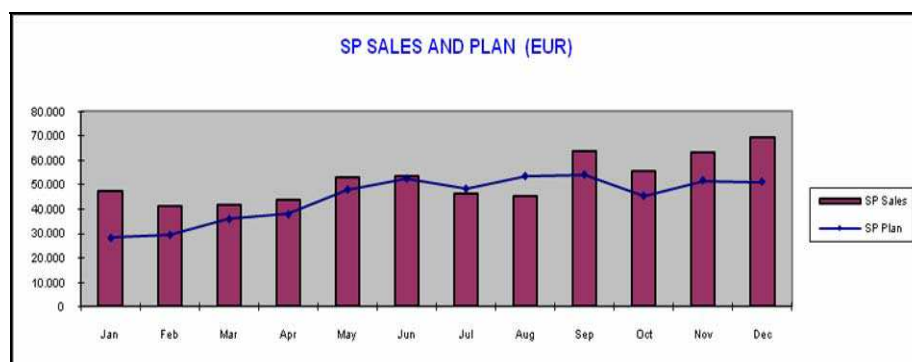


Figure 4. Spare parts turnover data

are Retail, Internal, and Warranty. In the case of independent body shop the labour sales mix may be split between Retail, Insurance, Franchised Dealers and Fleet. It is always wise to measure Labour Sales mix in Hours sold as opposed to the monetary value of Labour Sales because Recovery Rate vary across each income sector.

Operating profit

Operating profit = Operating Profit/Turnover (x100)

Operating profit is exactly the same as Departmental Profit and is calculated by taking Gross Profit minus Departmental Expenses. To make sense of this figure it is always expressed as a percentage of Turnover when used for trending as it is the direction of travel that is of most interest for analysis. Operating Profit should be more than 30%.

4.5. KPI in Business department

Absorption

Absorption = Aftersales direct profit /Total Overheads (x100)

The Absorption percentage of a business

essentially measures risk. How much of the company's overheads are covered by the profits generated by the Aftersales?

Manufacturers recommendations is that this factor should be even more than 100%.

Return on Sales (%)

Return on Sales = Net profit after interest/ Company Turnover (x100)

This is the KPI that many franchise manufacturers refer to when discussing levels of profitability within dealer networks.

It is also referred to as Net Profit After interest or Net Profit Before Tax on some financial reports and the only other item to be deducted from this figure is taxation.

It is important to monitor the trend of this KPI because a diminishing trend could indicate that dealership working harder for a lower return. Many business owners consider this to be one of the figures that holds the most importance, as it provides them with the profit dealer have retained as a percentage of the products and services that dealer have sold.

Return on Sales should be more than 2%.

Net profit after interest for a period of three year is shown on figure 6.

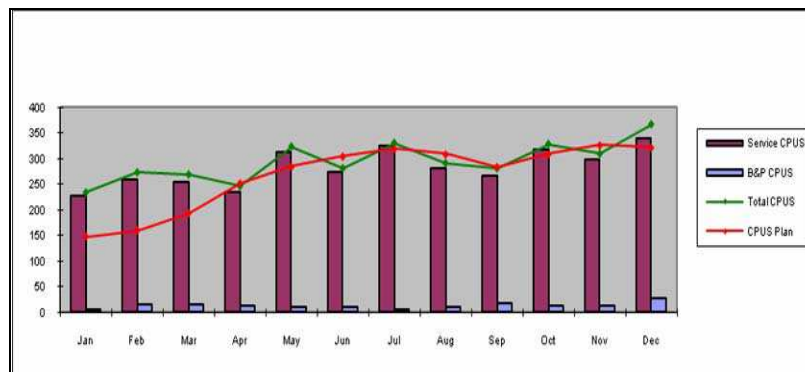


Figure 5. Body and Paint customer visits

5. CONCLUSION

Continual measuring of organizational performances via Key performance indicators is a newer concept that uses companies of today. KPI are financial and non – financial indicators that helps organization to testify how successful they are in their business. One of the necessary condition for effective and efficient system for performance measurement formulation is previously constituted process organization, with all standardized processes.

This paper gives explanation of some of the KPI that are used in cars dealerships. Graphical examples of KPI measured in Toyota dealerships, are also shown. Figures shows simplicity of measuring and explanation of a given results, what is one of

the most important advantage of using such indicators.

As it can be seen from those pictures measured values are always compared with some planed value, or benchmarked. Such review direct management whether they succeed in accomplishing set task. Also, through analysis over few years, it can be seen if organization accomplishes continual improvement of their business, what one of main principal of nowadays management systems is.

KPI performance measurement importance could also be expressed by next statement: „KPI tells you where performance has been in the past, where it is now, and perhaps more useful, where performance is likely to be in the future“ (Smith, 2001).

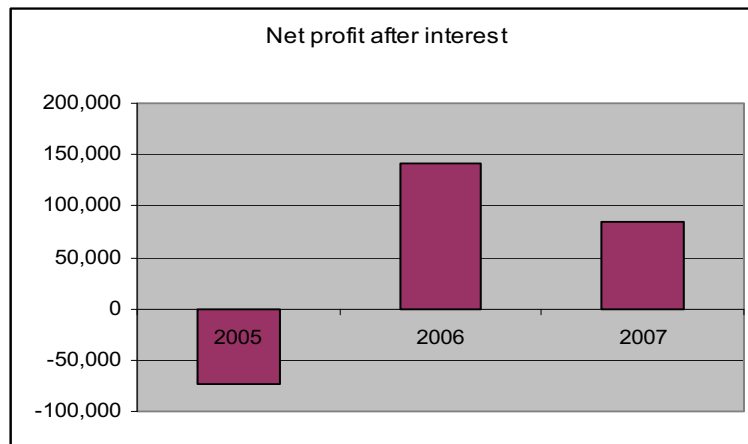


Figure 6. Net profit after interest

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УЛОГА И ЗНАЧАЈ МЕРЕЊА КЉУЧНИХ ИНДИКАТОРА ПЕРФОРМАНСИ (КРІ)

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Кључни индикатори перформанси (КРІ) су финансијски и нефинансијски показатељи које организације користе да би процениле и утврдиле колико су успешне, при чему се прати прогрес у правцу успостављених дугорочних циљева. Избор одговарајућих показатеља који ће се користити за мерење перформанси је изузетно важна активност. Да би одговарајући систем мерења укупног пословања организације преко КРІ био реализован, неопходно је претходно успоставити процесну организацију која укључује оријентацију на корисника и потребну флексибилност у данашњим условима глобалне конкуренције.

У раду је објашњен значај процесног приступа, зашто се мере КРІ и приказан практични пример мерења КРІ у Тојотиним аутосервисима.

Кључне речи: кључни индикатори перформанси (КРІ), процесни приступ, Тојотини аутосервиси